COEQUALIZER OF A PROPER GROUPOID ON A LOCALLY NOETHERIAN ALGEBRAIC SPACE EXISTS AND IS FINITE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2862-2880 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 2882-3015, the complete author proof (132 body lines).

Editorial note. Navigation stubs: 17 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 1 of queue_spaces-pushouts.json (pb=5812, stmt 2862-2880, proof 2882-3015).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-coequalizer' also exists in stacks-snapshot-20261011 — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00GQ: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3015–3015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GHL: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 450–469. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GG0: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 966–974. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 073K: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3015–3015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08AR: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3015–3015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A1B: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3015–3015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 081W: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3015–3015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03MV: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3015–3015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04WP: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3015–3015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03MH: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3015–3015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04XY: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3015–3015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03MG: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3015–3015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04ZK: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3015–3015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NX: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3015–3015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 040X: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3015–3015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 081X: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3015–3015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05Z1: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3015–3015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let*

$$g : Y \longrightarrow X$$

be a morphism of algebraic spaces over S. Assume X is locally Noetherian, and g is proper. Let $R = Y \times_X Y$ with projection morphisms $t, s : R \to Y$. There exists a coequalizer X' of $s, t : R \to Y$ in the category of algebraic spaces over S. Moreover

1. *The morphism $X' \to X$ is finite.*
2. *The morphism $Y \to X'$ is proper.*
3. *The morphism $Y \to X'$ is surjective.*
4. *The morphism $X' \to X$ is universally injective.*
5. *If g is surjective, the morphism $X' \to X$ is a universal homeomorphism.*

Proof. Denote $h : R \to X$ denote the composition of either s or t with g. Then h is proper by Morphisms of Spaces, Lemmas 04WP and 04XY. The sheaves

$$g_*\mathcal{O}_Y \quad \text{and} \quad h_*\mathcal{O}_R$$

are coherent $\mathcal{O}_X$-algebras by Cohomology of Spaces, Lemma 08AR. The X-morphisms s, t induce $\mathcal{O}_X$-algebra maps $s^\sharp, t^\sharp$ from the first to the second. Set

$$\mathcal{A} = \text{Equalizer}\left(s^\sharp, t^\sharp : g_*\mathcal{O}_Y \longrightarrow h_*\mathcal{O}_R\right)$$

Then $\mathcal{A}$ is a coherent $\mathcal{O}_X$-algebra and we can define

$$X' = \underline{\text{Spec}}_X(\mathcal{A})$$

as in Morphisms of Spaces, Definition 081W. By Morphisms of Spaces, Remark 081X and functoriality of the $\underline{\text{Spec}}$ construction there is a factorization

$$Y \longrightarrow X' \longrightarrow X$$

and the morphism $g' : Y \to X'$ equalizes s and t.

Before we show that X' is the coequalizer of s and t, we show that $Y \to X'$ and $X' \to X$ have the desired properties. Since $\mathcal{A}$ is a coherent $\mathcal{O}_X$-module it is clear that $X' \to X$ is a finite morphism of algebraic spaces. This proves (1). The morphism $Y \to X'$ is proper by Morphisms of Spaces, Lemma 04NX. This proves (2). Denote $Y \to Y' \to X$ with $Y' = \underline{\text{Spec}}_X(g_*\mathcal{O}_Y)$ the Stein factorization of g, see More on Morphisms of Spaces, Theorem 0A1B. Of course we obtain morphisms $Y \to Y' \to X' \to X$ fitting with the morphisms studied above. Since $\mathcal{O}_{X'} \subset g_*\mathcal{O}_Y$ is a finite extension we see that $Y' \to X'$ is finite and surjective. Some details omitted; hint: use Algebra, Lemma 00GQ and reduce to the affine case by étale localization. Since $Y \to Y'$ is surjective (with geometrically connected fibres) we conclude that $Y \to X'$ is surjective. This proves (3). To show that $X' \to X$ is universally injective, we have to show that $X' \to X' \times_X X'$ is surjective, see Morphisms of Spaces, Definition 03MV and Lemma 040X. Since $Y \to X'$ is surjective (see above) and since base changes and compositions of surjective morphisms are surjective by Morphisms of Spaces, Lemmas 03MH and 03MG we see that $Y \times_X Y \to X' \times_X X'$ is surjective. However, since $Y \to X'$ equalizes s and t, we see that $Y \times_X Y \to X' \times_X X'$ factors through $X' \to X' \times_X X'$ and we conclude this latter map is surjective. This proves (4). Finally, if g is surjective, then since g factors through $X' \to X$ we see that $X' \to X$ is surjective. Since a surjective, universally injective, finite morphism is a universal homeomorphism (because it is universally bijective and universally closed), this proves (5).

In the rest of the proof we show that $Y \to X'$ is the coequalizer of s and t in the category of algebraic spaces over S. Observe that X' is locally Noetherian (Morphisms of Spaces, Lemma 04ZK). Moreover, observe that $Y \times_{X'} Y \to Y \times_X Y$ is an isomorphism as $Y \to X'$ equalizes s and t (this is a categorical statement). Hence in order to prove the statement that $Y \to X'$ is the coequalizer of s and t, we may and do assume $X = X'$. In other words, $\mathcal{O}_X$ is the equalizer of the maps $s^\sharp, t^\sharp : g_*\mathcal{O}_Y \to h_*\mathcal{O}_R$.

Let $X_1 \to X$ be a flat morphism of algebraic spaces over S with X_1 locally Noetherian. Denote $g_1 : Y_1 \to X_1$, $h_1 : R_1 \to X_1$ and $s_1, t_1 : R_1 \to Y_1$ the base changes of g, h, s, t to X_1. Of course g_1 is proper and $R_1 = Y_1 \times_{X_1} Y_1$. Since we have flat base change for pushforward of quasi-coherent modules, Cohomology of Spaces, Lemma 073K, we see that $\mathcal{O}_{X_1}$ is the equalizer of the maps $s_1^\sharp, t_1^\sharp : g_{1,*}\mathcal{O}_{Y_1} \to h_{1,*}\mathcal{O}_{R_1}$. Hence all the assumptions we have are preserved by this base change.

At this point we are going to check conditions (1) and (2) of Lemma 0GHL. Condition (1) follows from Lemma 0GG0 and the fact that g is proper and surjective (because $X = X'$). To check condition (2), by the remarks on base change above, we reduce to the statement discussed and proved in the next paragraph.

Assume $S = \mathrm{Spec}(A)$ is an affine scheme, $X = X'$ is an affine scheme, and Z is an affine scheme over S. We have to show that

$$\mathrm{Mor}_S(X, Z) \longrightarrow \mathrm{Equalizer}(s, t : \mathrm{Mor}_S(Y, Z) \to \mathrm{Mor}_S(R, Z))$$

is bijective. However, this is clear from the fact that $X = X'$ which implies $\mathcal{O}_X$ is the equalizer of the maps $s^\sharp, t^\sharp : g_*\mathcal{O}_Y \to h_*\mathcal{O}_R$ which in turn implies

$$\Gamma(X, \mathcal{O}_X) = \mathrm{Equalizer}\left(s^\sharp, t^\sharp : \Gamma(Y, \mathcal{O}_Y) \to \Gamma(R, \mathcal{O}_R)\right)$$

Namely, we have

$$\mathrm{Mor}_S(X, Z) = \mathrm{Hom}_A(\Gamma(Z, \mathcal{O}_Z), \Gamma(X, \mathcal{O}_X))$$

and similarly for Y and R, see Properties of Spaces, Lemma 05Z1. □